

Khulna University of Engineering & Technology
Department of Urban and Regional Planning

BURP 1st Year 2nd Term Examination, 2020

URP 1211
(Urban Planning Principles)

Full Marks: 120

Time: 1:30 hrs

- N.B.** i) Answer any two questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable value for any missing data.

Section – A

1. (a) What characteristics make the Town Centre a “Social Hub”? List some examples that help differentiate between amenities and utilities. (15)
(b) Are urban planning principles across geographic domain as exact science? – Explain. Write down the guiding principles of Town Planning and the characteristics for successful Urban Planning. (15)
2. (a) Describe the different types of town growth with proper examples and diagrams. (15)
(b) Classify recreational area with proper examples. Discuss the existing condition of open space in Dhaka city and the methods to overcome the challenges. (15)
3. (a) Illustrate the concept of Garden city with the help of diagrams. (15)
(b) Briefly describe the priorities of the different types of users in a railway station. (10)
(c) List some requirements for passengers and bus operators in a bus terminal station. (05)

Section – B

4. (a) Classify the towns according to their physical pattern with necessary diagrams. (15)
(b) Write the assumptions of “Central Place Theory”. Also illustrate the three principles in the arrangement of the theory. (15)
5. (a) Describe the functional classification of roads with perspective of some common attributes. (15)
(b) Describe the impacts of industrial parks and possible solution to reduce it through eco-industrial park. (15)
6. (a) Illustrate the industrial classification according to nature of processing, types of products and locational characteristics. (15)
(b) Define the sea port development. Diagrammatically describe the runway orientation of airport. (15)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 2nd Term Regular Examination, 2020
URP 1251
(Introduction to History of Architecture)

Full Marks: 120

Time: 1.5 hrs

- N.B.** i) Figures in the right margin indicate full marks.
ii) Assume reasonable value for any missing data.
iii) Answer any two questions from each section in separate script.

Section – A

- 1(a) Define "History of Architecture" according to your own understanding. Briefly describe the factors that influence architectural development from ancient period to contemporary period. (20)
- 1(b) Write short notes on (any two): (10)
- (i) Royal pyramid (ii) Notre-Dam De Paris (iii) Rib vault
- 2(a) Describe the master plan of the Greck acropolis with proper illustration. (15)
- 2(b) "Organic Architecture shaped the ideology of Frank Lloyd Wright"-describe the statement with necessary examples. (15)
- 3(a) Illustrate the transition features from Romanesque to Gothic Architecture with relevant sketches. (15)
- 3(b) Elaborate the five points of Le Corbusier which shaped the modern Architecture. (15)

Section – B

- 4(a) Traditional bengal feudal palaces has specific zoning, space organization and design criteria influenced by Occidental and Oriental Architecture".- describe this statement with diagrams. (20)
- 4(b) Describe any two of Indo-Islamic architectural styles with necessary sketches. (10)
- 5(a) "Modern Architecture concept came from the realization that the future should not hinge on the past"- Illustrate it with Expressionism, Cubism, De Stijl and Bauhaus concept. (15)
- 5(b) Suppose, you are engaged in a Science Academy development project of a particular urban central "X". Which principles you will follow to achieve sustainable architecture for this project and why? (15)
- 6(a) Write short notes (any five): (30)
- (i) Philosophical interpretation of Vihara (ii) Rekha Deul
(iii) Mughal architectural features (iv) Problems of "ethical" brutal approach
(v) Optical correction (vi) Features of Bungalow (vii) Basic elements of temple

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1stYear 2nd Term Regular Examination, 2020
Math 1217
(Mathematics-II)

Full Marks: 120

Time: 1.5 hrs

- N.B.** i) Figures in the right margin indicate full marks.
ii) Assume reasonable value for any missing data.
iii) Answer any two questions from each section in separate script.

Section – A

- 1 (a) Discuss the continuity and differentiability of $f(x)$ at $x=1$, Where 12

$$f(x) = x ; \text{ when } 0 < x < 1$$

$$= 2-x ; \text{ when } x \geq 1$$
- (b) Find $\frac{dy}{dx}$, where $y = x^{\cos^{-1}x} + e^{\frac{x}{2}-3}$ 08
- (c) Determine the n^{th} derivative of $y = \frac{x}{x^2-1}$ 10
- 2 (a) State Leibnitz's theorem. If $y = \sin^{-1}x$, then find y_{n+2} by using Leibnitz's theorem. 15
- (b) Discuss the maxima and minima of $4x+9y$, when $xy=4$ 15
- 3 (a) Find the general solutions of $\frac{d^2y}{dx^2} - 4y = e^{2x} - \sin x$ 15
- (b) Solve $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = x^3e^x + 5$ 15

Section – B

- 4 a) Calculate $\int \frac{dx}{(x^2+3)\sqrt{x^2+4}}$ 15
- b) Evaluate $\int_0^{\pi/2} \frac{dx}{(a^2 \cos^2 x + b^2 \sin^2 x)^2}$ 15
- 5 (a) If $I_n = \int_0^{\pi/4} \tan^n \theta d\theta$ then show that, $I_n = \frac{1}{n-1} - I_{n-2}$ and hence find the value of $\int_0^{\pi/4} \tan^6 \theta d\theta$ 15
- (b) Find the area bounded by the asteroid $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$. 15
- 6 (a) Determine the constant A such that the equation $(x^2 + 3xy)dx + (Ax^2 + 4y)dy = 0$ is exact and solve the resulting exact equation 15
- (b) Solve the initial value problem: 15

$$\frac{dy}{dx} + \frac{y}{2x} = \frac{x}{y^3}; y(1)=2$$

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HUM 1217
(Principles of Economics)

Full Marks: 120

Time: 1.5 hrs

- N.B.** i) Answer any TWO questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Why is “what to produce” a problem in every economy? How does the price mechanism solve this problem in a centralized economy? (15)
- (b) The values in the following table refer to the change in an individual consumer's consumption of coffee and tea at home when the price of coffee rises (everything else, including the price of tea remains the same). (10)
- Draw a figure showing the changes, and
 - Explain the figure drawn.

	Before		After	
	Price (cents/cup)	Quantity (cups/month)	Price (cents/cup)	Quantity (cups/month)
Coffee	40	50	60	30
Tea	20	40	20	50

- (c) Mention why demand curve left to right downward? (05)
2. (a) What does the price elasticity of demand measure in general? Is the price elasticity for white bread greater than the price elasticity for bread in general? Why? What general rule can we infer from this? (10)
- (b) What factors govern the size of the coefficient of price elasticity of demand? (15)
- (c) What is meant by production function? Give an example. (05)
3. (a) How does a firm in a perfectly competitive market find the best level of output? (15)
- (b) How does a firm make sure that the firm is at risk position? Draw necessary figures and explain it. (15)

Section – B

4. (a) Distinguish between GNP and GDP. (10)
- (b) Explain the relationship between GDP and economic wellbeing. (20)
5. (a) What is demand-pull and cost-push inflation? (10)
- (b) Discuss causes of inflation. Give your suggestions to control of inflation. (20)
6. (a) What is meant by investment multiplier and on what factor does it depend? (10)
- (b) Explain how commercial banks create credit. (20)

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URP 1281

(Surveying and Cartography)

Full Marks: 120

Time: 1:30hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable value for any missing data.

Section – A

1. (a) Suppose a large river interrupts the chain line. Describe the method to overcome the obstacle during chain survey. (15)
(b) The followings are the bearings of a closed traverse: (15)

<u>Side</u>	<u>FB</u>	<u>BB</u>
AB	N 45° 30' E	S 45° 30' W
BC	S 55° 0' E	N 65° 0' W
CD	S 10° 30' W	N 10° 30' E
DA	N 70° 45' W	S 80° 45' E

Calculate the interior angles of the traverse.

2. (a) The following consecutive readings were taken with a dumpy level along a chain line at a common interval of 20m. The first reading was at a chainage of 175m where the RL is 97.075. The instrument was shifted after the fifth and ninth readings.
3.250, 2.255, 1.825, 0.950, 3.225, 2.760, 2.135, 1.470, 1.965, 1.825, 2.190 and 2.935 m.
Make rules on a page of your notebook in the form of level book page and enter on it. The above readings and find the RL of all points by any of one method.
(b) What do you mean by reconnaissance survey? (05)
3. (a) Illustrate the method of intersection of plane tabling with necessary figure. (15)
(b) Illustrate the use of contour map. (05)
(c) The following offsets were taken at 20m intervals from a survey line to an irregular boundary line. (10)
3.25, 3.50, 4.75, 6.85, 5.50, 7.80, 8.90, 5.35, 4.25, 4.50m
Calculate the area enclosed between the survey line, the irregular boundary line and the first and last offsets by-
a) The Trapezoidal rule
b) Simpson's rule

Section – B

4. (a) Describe the general classification of maps with appropriate example. (25)
(b) What are the basic elements of map composition? (05)
5. (a) Describe different ways to indicate map scale. (06)
(b) Write short notes (any three)- (3x8)
i. Graduated Symbol map
ii. Cadastral map

- iii. Symbology map legend
- iv. RF converting

6. (a) Demonstrate the Ellipsoid and Geoid earth concept with necessary geometrical indications. (20)
- (b) Define the association of longitude and time zone measurement with necessary figures. (10)